

Program No. 7

Aim

To implement TCP-based client-server communication in which the client accepts an integer N, generates an array of size N with random numbers in the range 1 to 50, sends the array to the server, and the server calculates the sum and average of the array and returns the result to the client for display.

Algorithm - Client

1. Start.
2. Create a TCP socket.
3. Connect to the server using its IP address and port number.
4. Read the integer N from the user.
5. Generate an array of N random numbers in the range 1 to 50.
6. Convert the array into a comma-separated string.
7. Send the array to the server.
8. Receive the sum and average from the server.
9. Display the received result.
10. Close the socket.
11. Stop.

Algorithm - Server

1. Start.
2. Create a TCP socket.
3. Bind the socket to the specified IP address and port number.
4. Listen for incoming client connections.
5. Accept the client connection.
6. Receive the array from the client.
7. Convert the received string into an integer array.
8. Calculate the sum of the array elements.

9. Calculate the average of the array elements.
10. Send the sum and average back to the client.
11. Close the client connection and server socket.
12. Stop.